

Foundations of AI for Business: Introducing the Course

What this course is about: **foundations** of AI

- You will learn around **12-15 foundational AI ideas** relevant to your career.

Exercise (2 min.) | Let us generate our syllabus using LLMs

- Work with a partner. Prompt your favorite LLM:

*“Give us a list of the 12 or so foundational AI ideas to include in an introductory course on AI”.**

*You can vary the language of the prompt a bit.

“Some people ask why **no laptops** and **no phones**. The answer is, we humans only have one language processor. If your language processor is engaged, **you’re distracted**. Worse, you **distract all of the people around you**.”

Patrick Winston (Developed and taught MIT’s Intro AI course for many years)



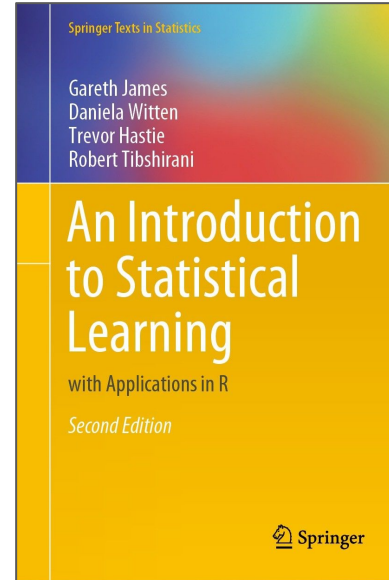
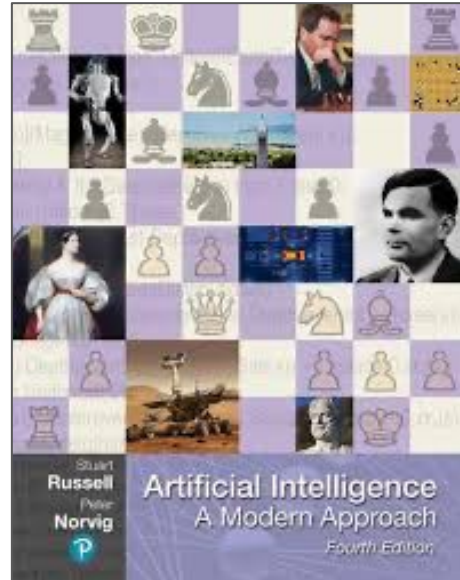
Goals of the course |

- Learn to **think critically** about foundational AI ideas, to empower you to **develop** your own AI ideas that:
 - (1) **reflect real-world phenomena** you care about;
 - (2) are **based on AI foundations**, beyond fashions
 - (3) are **actionable** as AI systems you can design or participate in

Motivation for me to develop this course |

- No genuinely foundational intro. to the AI field exists that I know of (either in mgmt. or computer science schools).

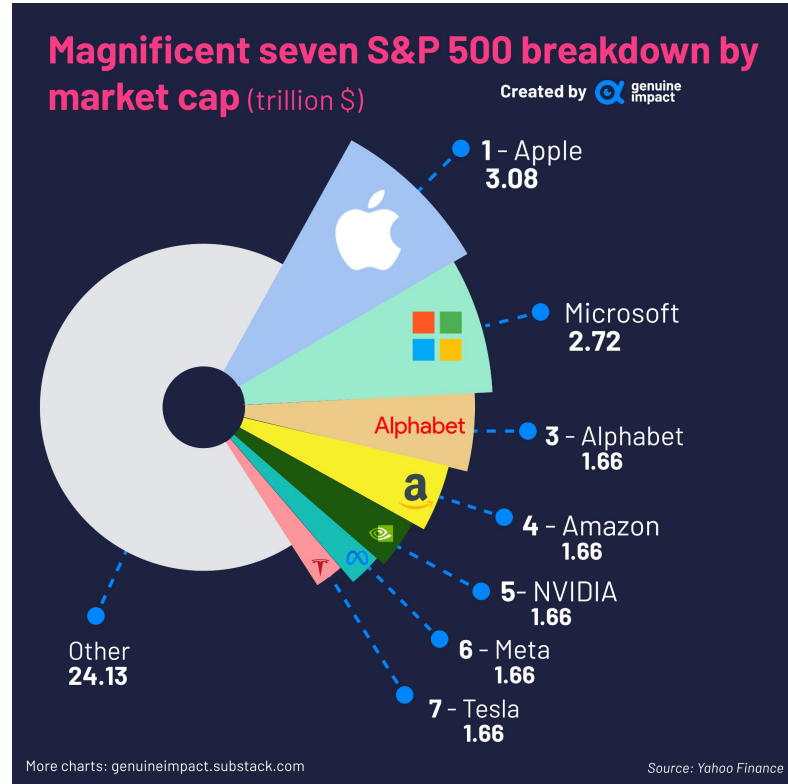
Most AI
textbooks:
**toolkits of
algorithms**



Last 10-15
years or so:
toolkit narrows
to only
**machine
learning
algorithms**

Currently-popular view of AI | **AI = machine learning**

Until a few years ago, this was understandable...



Emerging view of AI | AI = ML + many other things

In a GenAI era, “AI = machine learning” is not close to good enough to define AI



Emerging view of AI | AI = ML + many other things

INNOVATION > AI

Will LLMs Become Obsolete?

By [John Werner](#), Contributor. ⓘ I am an MIT Senior Fellow & Lecturer, 5x-foun... [Follow Author](#)

Published Feb 14, 2026 at 01:18pm EST

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🌻 **claude code psychosis**

are your problems software-shaped?

 JASMINE SUN
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If you tell a friend they can now instantly create any app, they'll probably say "Cool! Now I need to think of an idea." Then they will forget about it, and never build a thing. The problem is not that your friend is horribly uncreative. It's that most people's problems are not software-shaped, and most won't notice even when they are.

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 **Matt Shumer** ✓ @mattshumer_ · 20h [🔗](#) [...](#)

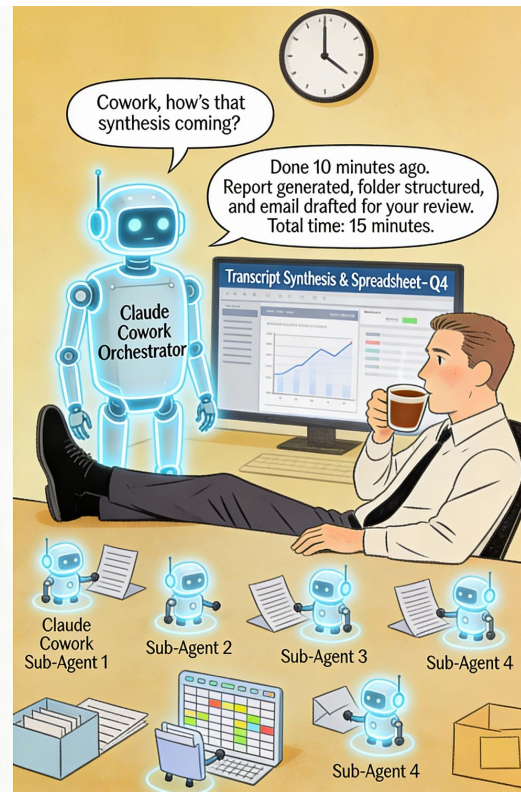
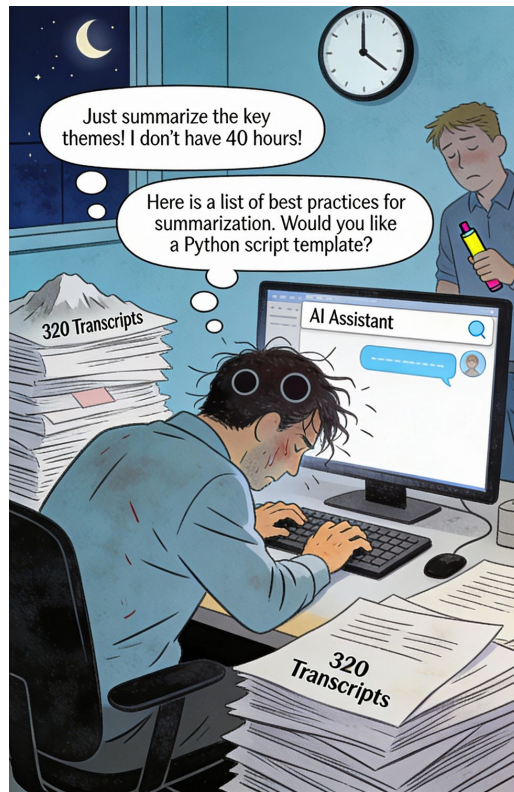


📄 Article

Something Big Is Happening

Think back to February 2020. If you were paying close attention, you might have noticed a few people talking about a virus spreading overseas. But most of us weren't paying close attention. The stock...

Emerging view of AI |
AI = reasoning,
planning, coordination,
communication,
managing... etc.



Where AI seems to be headed

- **Technology:** diverse approaches beyond machine learning (reasoning, coordination, planning, etc.)
- **Skills you will need: holistic problem-solving**, not just coding/data science skills.
- ***Implication: you will need to study diverse AI foundations (this course!)***

To study, diverse AI foundations you will probably need to **unlearn** some popular views of AI. Let us first establish:

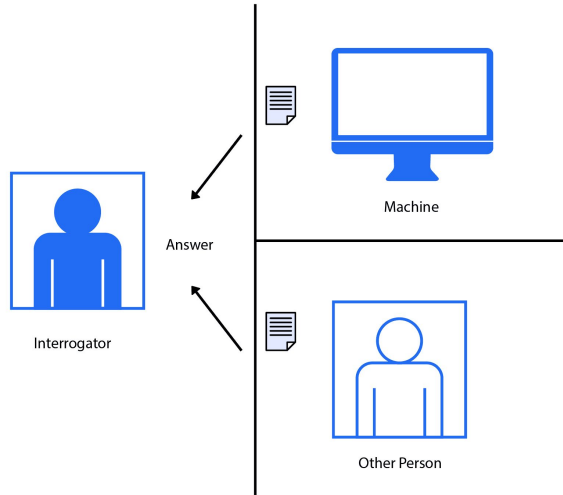
- (1) How we will **define AI** in this course
- (2) What we mean by **foundational AI ideas**
- (3) How we will view **AI today**

1. Defining AI

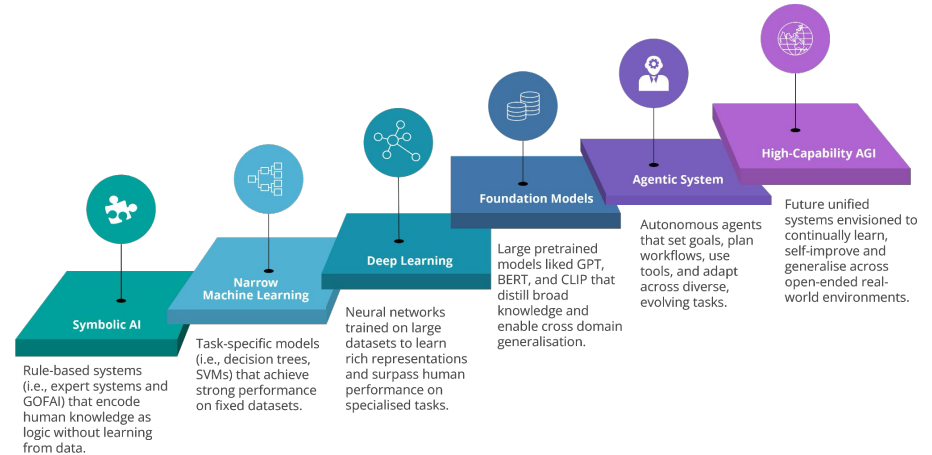
- How will we define AI for the purposes of this course?

Defining AI | One option: comparison to humans

- AI = “making a machine behave in ways that, if a human did them, would be considered intelligent” (Dartmouth conference 1956)



Turing Test (1950)



Artificial General Intelligence (AGI) (2027?!)

Defining AI | This course: **NO** comparison to humans

- Our approach: AI = diverse intelligences for learning to solve **complex problems** arising in **real-world phenomena**



Not robot vs. human chef



Instead: cooking as a phenomenon, what system of intelligences are needed?

Why not compare AI to human intelligence? So we can focus on our goal of introducing AI foundations

- (1) Encourage you in this course to **compare diverse AI ideas** as they relate to a complex problem.
- (2) Discourage you in this course from asking questions like: “*can AI be conscious?*”, “*will AI automate all our jobs*”, which require **first** studying AI foundations

AI, for us, equals:

- Some **artificial system**
- The system **learns** to solve complex problems
- AI = the **system itself**, or **ideas** about how to design it

Rules for how we will define AI in this course

1. We will define AI in this course as any **artificial system** designed to **learn to solve complex problems** about real-world phenomena.
2. We may define AI in terms of both **diverse technologies** and **ideas**.
3. We will not talk of whether AI is conscious, autonomous, or otherwise human-like or superhuman, unless there is explicit reason to do so.
4. Whenever we use the term “AI” by itself, it is always to refer to **the full set of diverse ideas or technologies** developed in the AI field since its origins.

2. What do we mean by “foundational” AI ideas

- **Consider first** — what if we had defined AI by comparing to humans? If so, then:
 - Same problems as **psychology, neuroscience**, etc.
 - AI = simulating human mental processes as **algorithms**
 - Comp. sci. version of this: create “**optimal**” algorithms

2. What do we mean by “foundational” AI ideas

- **Our approach** — AI as artificial systems designed to learn to solve complex problems arising in real-world phenomena.
 - Same problems as **management / organization theory**
 - AI = **systems of diverse elements** (*algorithms, humans, digital computers, analog devices, equipment, facilities, other animate or inanimate objects*).

Our fundamental analogy:

AI and organizations are the same thing

- **Consider:** what would you have to change in our AI definition (“*artificial systems designed to learn to solve complex problems*”) to be consistent with a definition of **organizational intelligence**?
- Our point of view will be “**nothing**”. **AI and orgs. = direct analogs**

Foundational AI ideas
= organizational
theories, and vice
versa

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Organizations as Artificial Intelligences: The Use of Artificial Intelligence Analogies in Organization Theory

Felipe A. Csaszar and Tom Steinberger

Published Online: 26 Jan 2022 | <https://doi.org/10.5465/annals.2020.0192>

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Abstract

A rarely acknowledged fact about organization theory (OT) is that many of its ideas stem from the field of artificial intelligence (AI). For example, key OT concepts such as problemistic search, heuristics, exploration, requisite variety, and organizational scripts all have their roots in AI. The main goal of this paper is to expose the full range of AI ideas that have been used in OT. We do so by explaining key AI ideas and showing how OT used them. Our review covers over 100 OT works that depend on AI ideas both critically and explicitly. We group these ideas into 10 AI approaches that speak to three fundamental processes in organizations: search, representation,

My graduate school
department chair's
advisor: **Herbert
Simon** — Pioneer of
the AI field and
modern management
schools



1978 Nobel Prize in Economics
and
1976 Turing Award (the Nobel for computer
scientists)

Rules for how we will define AI in this course

5. *AI and organizations are the **same thing** — artificial systems designed to learn to solve complex problems — and can be studied using the **same foundational ideas**.*

6. *Unlike organizations, algorithms are neither examples of, nor direct analogs to, AI technologies or ideas. We will avoid using the term “algorithm” in this course altogether, except when there is a specific reason to do so.*

Overview of the course

- Content: 12-15 AI ideas and analogs in organizations
- Structure: Monday lecture, Wednesday exercises
- Grading:
 - Weekly short quizzes (10x6%) (can drop one quiz)
 - Group project (20%)
 - Participation (20%) (including attendance!)